MOMENT MAPS

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1. Some Symplectic Geometry

We will work with complex manifolds – given a complex manifold X, we will denote its holomorphic tangent and cotangent bundles by TX and T^*X respectively.

Definition 1.1. Let X be a complex manifold. A ***holomorphic symplectic structure*** on X is a closed, nondegenerate holomorphic 2-form $\omega \in H^0(X, \Lambda^2 T^*X)$.

Example 1.2. The prototypical example of a holomorphic symplectic manifold is the holomorphic cotangent bundle T^*X. If we let $\{p^i\}$ be local holomorphic coordinates for X, we get a local trivialization of T^*X using the one forms dz^i. This then gives holomorphic coordinates $\{p^i, q^i\}$ on T^*X where given a cotangent vector ξ, we have

$$\xi = \sum_i q^i(\xi) dz^i$$

One can then check that the two form locally defined by

$$\omega := \sum_i dp^i \wedge dq^i$$

defines a symplectic structure on T^*X.

For a more coordinate free description, let $\pi : T^*X \to X$ be the projection map. Then the ***tautological one form*** is the one form λ on T^*X defined by

$$\lambda_{(x,\xi)} = d\pi_x^* \xi$$

One can then check that the symplectic form ω above is the same as $d\lambda$.

Definition 1.3. Let (X_1, ω_1) and (X_2, ω_2) be holomorphic symplectic manifolds. A map $F : X_1 \to X_2$ is a ***symplectomorphism*** if it is an isomorphism and $F^*\omega_2 = \omega_1$. A group action of G on (X, ω) is ***symplectic*** if G acts by symplectomorphisms.

Definition 1.4. A vector field V on (X, ω) is ***symplectic*** if the Lie derivative satisfies $\mathcal{L}_V \omega = 0$. Alternatively, the flow of V acts by symplectomorphisms.

Using the symplectic form ω, we can construct many symplectic vector fields out of functions on X. Since ω is nondegenerate, we get an isomorphism

$$\begin{aligned} \tilde{\omega} : TX &\to T^*X \\ V &\mapsto \omega(V, \cdot) \end{aligned}$$

Then given a function f, we can construct the vector field $V_f := \tilde{\omega}^{-1}(df)$. This also endows the ring of functions $\mathcal{O}_X$ on X with a Lie algebra structure.

Definition 1.5. Let f, g be functions on X. Then the ***Poisson bracket*** of f and g, denoted $\{f, g\}$ is the function

$$\{f, g\} := \omega(V_f, V_g)$$

Now suppose we have group G acting symplectically on X, and let $\mathfrak{g}$ be its Lie algebra. Then given a Lie algebra element $a \in \mathfrak{g}$, we get a vector field on X whose value at $p \in X$ is

$$\left.\frac{d}{dt}\right|_{t=0} \exp(ta) \cdot p$$

One can check that this defines a symplectic vector field, and that this map $\mathfrak{g} \to \Gamma(X, TX)$ is Lie algebra homomorphism.

Definition 1.6. A symplectic action of G on X is ***Hamiltonian*** if there exists a Lie algebra homomorphism $H : \mathfrak{g} \to \mathcal{O}_X$ such that the following diagram commutes:

$$\begin{tikzcd} & \mathfrak{g} \arrow[dl, "H"'] \arrow[dr] & \\ \mathcal{O}_X \arrow[rr] & & \Gamma(X, TX) \end{tikzcd}$$

The function H is called a ***Hamiltonian*** of the action, and we denote $H(a) = H_a$.

Definition 1.7. Given a Hamiltonian action of G on X with Hamiltonian H, the ***moment map*** of the action is the equivariant map $\mu : X \to \mathfrak{g}^*$, where given $x \in X$, the linear functional $\mu(x)$ is the one that sends $a \in \mathfrak{g}$ to $H_a(x)$.

Another way to interpret a moment map $\mu : X \to \mathfrak{g}^*$ is as follows: given an element $a \in \mathfrak{g}$, we can interpret it as an element of the double dual $(\mathfrak{g}^*)^*$. We can pullback functions on $\mathfrak{g}^*$ to functions on X along μ, and we get that $H_a = \mu^* a$.

Example 1.8. Suppose we have an action of G on X. Then we get an induced action on T^*X where $g \cdot (x, \xi) = (g \cdot x, (dg_x^{-1})^* \xi)$. This action is symplectic with respect to the symplectic form $\omega = d\lambda$, where λ is the tautological one form on T^*X.

In fact, this action is Hamiltonian. To see this, we first note that we can lift vector fields on X to vector fields on T^*X. In local coordinates $\{p^i\}$ for X, let $\{p^i, q^i\}$ be the corresponding coordinates on T^*X. Then for any vector field $U = f^i \partial/\partial p^i$ on X, we can define a vector field $\tilde{U}$ on T^*X by the same formula. One can then check that the Hamiltonian of this action is given by $H_a = \lambda(\widetilde{V_a})$.

We now connect this to the discussion of differential operators D_X. In a slight abuse of notation, we let D_X denote the ring of global diferential operators on X. Suppose we have

an action of G on X, which gives us a Hamiltonian action of G on T^*X with moment map $\mu : T^*X \to \mathfrak{g}^*$. The infinitesimal action of G on X gives a map from $\mathfrak{g}$ to vector fields on X, which are derivations on functions on X, i.e. differential operators of order one. The map $\mathfrak{g} \to \Gamma(X, TX)$ given by the action then extends to an algebra map $\mathcal{U}\mathfrak{g} \to D_X$, where $\mathcal{U}\mathfrak{g}$ is the universal enveloping algebra of $\mathfrak{g}$. The universal enveloping algebra $\mathcal{U}\mathfrak{g}$ also has a filtration, where the pth filtered piece consists of the span of words of length less that p in the elements of $\mathfrak{g}$. The map $\mathcal{U}\mathfrak{g} \to D_X$ respects the filtrations on both sides, so it induces a map $\mathrm{gr}(\mathcal{U}\mathfrak{g}) \to \mathrm{gr}(D_X)$ of the corresponding associated graded algebras. We then recall that we have isomorphisms $\mathrm{gr}(\mathcal{U}\mathfrak{g}) \cong \mathrm{Sym}(\mathfrak{g})$ and $\mathrm{gr}(D_X) \cong \mathrm{Sym}(TX)$.

Proposition 1.9. *The induced map* $\mathrm{Sym}(\mathfrak{g}) \to \mathrm{Sym}(TX)$ *is the same as the map* μ^*.

Proof. it suffices to check that both maps agree on the elements $a \in \mathfrak{g}$, thought of as elements of the double dual of $\mathfrak{g}$. As mentioned earlier, we have that $\mu^* a = H_a$. On the other hand, the image of a under the map $\mathrm{Sym}(\mathfrak{g}) \to \mathrm{Sym}(TX)$ is the vector field $\widetilde{V_a}$ obtained by lifting the infinitesimal action of a to T^*X. The desired equality then follows from the fact that the Hamiltonian H makes the following diagram commute:

$$\begin{tikzcd} & \mathfrak{g} \arrow[dl, "H"'] \arrow[dr, "\widetilde{V}"] & \\ \mathrm{Sym}(TX) \arrow[rr] & & \Gamma(T^*X, T(T^*X)) \end{tikzcd}$$

where we use the fact that the relative Spec of $\mathrm{Sym}(TX)$ is T^*X. ■

From this perspective, one can think of the map $\mathcal{U}\mathfrak{g} \to D_X$ as a noncommutative "quantization" (thought I don't know what that word means) of the ordinary moment map $\mu : T^*X \to \mathfrak{g}^*$. One obtains the pullback map μ^* by "abelianizing" the map $\mathrm{U}\mathfrak{g} \to D_X$ by passing to the associated graded.

2. The Springer Resolution

Let G be a semisimple group over $\mathbb{C}$ – for concreteness it does very little harm to assume that $G = \mathrm{SL}_3\mathbb{C}$. Then fix a maximal torus T (with Lie algebra $\mathfrak{h}$) and a Borel subgroup B (with Lie algebra $\mathfrak{b}$) containing T. Finally, we let $\langle \cdot, \cdot \rangle$ denote the killing form. In the $\mathrm{SL}_3\mathbb{C}$ case, we can choose these to be:

(1) T is the subgroup of determinant 1 diagonal matrices, and $\mathfrak{h}$ is the subalgebra of trace zero diagonal matrices.
(2) B is the subgroup of determinant one upper triangular matrices, and $\mathfrak{b}$ is the subalgebra of trace zero upper triangular matrices.

A ***flag variety*** is then a variety isomorphic to the homogeneous space G/B. The quotient map $G \to G/B$ then has the structure of a principal B-bundle.

Proposition 2.1. *We have equivariant vector bundle isomorphisms:*

(1) $T(G/B) \cong G \times^B \mathfrak{g}/\mathfrak{b}$
(2) $T^*(G/B) \cong G \times^B \mathfrak{b}^\perp$, *where the orthogonal complement is with respect to the Killing form* $\langle \cdot, \cdot \rangle$.

Proof. We have a surjective vector bundle map $G/B \times \mathfrak{g} \to T(G/B)$ given by sending (gB, x) to the evaluating of the vector field V_x given by the infinitesimal action of G on

G/B at the point gB. The kernel E of this map is the vector bundle whose fiber over gB is the Lie algebra of the stabilizer of gB, which is a Borel subalgebra gBg^{-1}. We then have an isomorphism $G \times^B \mathfrak{b} \to E$ given by sending the equivalence class $[g, x]$ to (gB, gxg^{-1}). Therefore, we get an isomorphism $T(G/B) \cong (G/B \times \mathfrak{g})/(G \times^B \mathfrak{b}) \cong G \times^B \mathfrak{g}/\mathfrak{b}$. Since the Killing form is nondegenerate, we can use it to identify the dual $(\mathfrak{g}/\mathfrak{b})^*$ with $\mathfrak{b}^\perp$, which then gives us that $T^*(G/B) \cong G \times^B \mathfrak{b}^\perp$. ■

The space $\mathfrak{b}^\perp$ also has another description. Let

$$\mathfrak{g} = \bigoplus_\alpha \mathfrak{g}_\alpha$$

be the root space decomposition. Then we get a direct sum decomposition $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}$, where $\mathfrak{n}$ is the direct sum of the root spaces of the roots in the chamber determined by our choice of Borel subalgebra $\mathfrak{b}$, and consists of the nilpotent elements in $\mathfrak{b}$. We then claim that $\mathfrak{b}^\perp = \mathfrak{n}$. To see this, it suffices to show the containment $\mathfrak{n} \subset \mathfrak{b}^\perp$ by dimension considerations. We then note that for $b \in \mathfrak{b}$ and $n \in \mathfrak{n}$, we have $\langle b, n \rangle = \operatorname{tr}(\operatorname{ad}(b) \operatorname{ad}(n))$. We then nonte that $\operatorname{ad}(b) \operatorname{ad}(n)$ cannot have any nonzero eigenvectors, since n can be written as a sum of elements of positive root spaces, and b can be writen as a sum of elements of nonnegative root spaces. Therefore, $\operatorname{ad}(b) \operatorname{ad}(n)$ necessarily moves any $\mathfrak{g}_\alpha$ to a different root space. Therefore, $\langle b, n \rangle = 0$.

Since G acts on G/B, we get a Hamiltonian action of G on $T^*(G/B) \cong G \times^B \mathfrak{n}$. We then want to identify the moment map as a map $\mu : G \times^B \mathfrak{n} \to \mathfrak{g}^*$. We recall that for $a \in \mathfrak{g}$, the Hamiltonian H_a is the function $\lambda(\widetilde{V_a})$, where λ is the tautological one form on $T^*(G/B)$. Therefore, given $[g, x] \in G \times^B \mathfrak{n}$ and $a \in \mathfrak{g}$, we compute

$$\begin{aligned} H_a([g, x]) &= \lambda_{[g,x]}(\widetilde{V_a}|_{[g,x]}) \\ &= \pi^*(gxg^{-1})(\widetilde{V_a}|_{[g,x]}) \\ &= \langle gxg^{-1}, a \rangle \end{aligned}$$

Therefore, using the Killing form to identify $\mathfrak{g}$ with $\mathfrak{g}^*$, the moment map is given by the simple formula $\mu([g, x]) = gxg^{-1}$. Using the identification $T^*(G/B) \cong G \times^B \mathfrak{n}$, we get a map $T^*(G/B) \to \mathcal{N}$, where $\mathcal{N} \subset \mathfrak{g}$ is the ***nilpotent cone*** – the subvariety of nilpotent elements in $\mathfrak{g}$. The map is given by $[g, x] \mapsto gxg^{-1}$, which is exactly the moment map of the action of G on $T^*(G/B)$, and is called the ***Springer resolution***.

The Springer resolution plays a role in the proof of Beilinson-Bernstein localization, which states that taking global sections gives an equivalence between D-modules on G/B and certain representations of $\mathfrak{g}$. One step of this involves computing what the ring D_X is, which is made possible by passing to the Springer resolution by taking associated gradeds.